Communicating with Data – Persuasive Vis

Dr. Ab Mosca (they/them)

Designing Visual Stories & Graphics

MONDAY
NOV 2
KATHLEEN MCCARTNEY HALL

6–7PM KMH 106 Panel Conversation

Moderated by **Ab Mosca,** Visiting Assistant Professor in Computer Science and Statistical & Data Sciences





Francesca Paris is a reporter for The Upshot, a section of *The New York Times* specializing in explanatory, analytical, and visual journalism.



Tanvi Sharma is a design technologist working at the intersection of data, aesthetics, and the humanities.



Ally Levine writes, illustrates and develops visually-driven stories on the Reuters Graphics Desk.

7:15–8:15PM Interactive Workshops

Draw Your Own Chart

Pick a topic, find data, look at some different chart forms and then use paper, pens, tape, and glue to make physical representations of the numbers.

KMH 101

Visualizing Histories: A Peabody-Inspired Data Workshop

Drawing from your own calendar, select a timeframe — maybe a month — to document events, asking: what do we remember, forget, or want to forget?

KMH 106

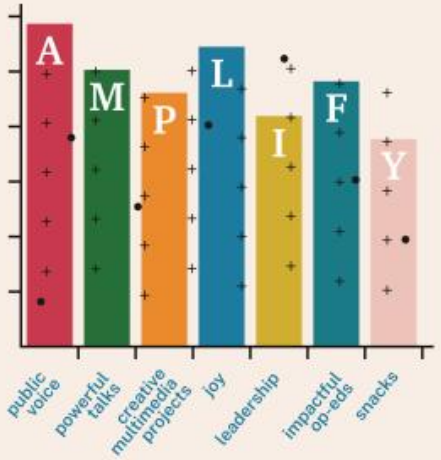
Hey Data, Make Me Feel Something

Go beyond simply showing the trend and explore how to visually represent data in ways that evoke the feeling of the story.

KMH 201



Smith College Wurtele Center for Collaborative Leadership



SMITH COLLEGE
Wurtele Center
for Collaborative Leadership

What is Amplify?

Amplify is a program for Smith students to gain the skills, coaching, and platform you need to share your knowledge and perspectives with a public audience.

Events and workshops focus on practices of public voice including public writing, speaking, and other creative mediums.

The program culminates with a chance to participate in the Amplify Competition, which is open to all Smith undergraduates.

Get Inspired

MONDAY 11/2 IN MCCARTNEY HALL

6–7PM: PANEL CONVERSATION

Designing Visual Stories and Graphics

Moderated by **Ab Mosca**, Visiting Assistant Professor in Computer Science and Statistical & Data Sciences



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7:15–8:15PM: WORKSHOPS

- *Draw Your Own Chart* — McCartney Hall 101
- *Visualizing Histories: A Peabody-Inspired Data Workshop* — McCartney Hall 106
- *Hey Data, Make Me Feel Something* — McCartney Hall 201

MONDAY 11/9

Registration and Submission Forms open for the Amplify Competition

Create & Submit

EARLY JANUARY

Prepare a submission for Amplify! Write an op-ed, develop a public talk, or design a visual story or graphic

DURING J-TERM

Get some feedback or coaching on your Amplify submission from Wurtele Center staff

WEDNESDAY 1/20

Last day to register for the Public Speaking Competition and submit to the Public Writing and Wild Card categories

SATURDAY 1/30

Registered Public Speakers attend workshop with Emily Brown '11, Public Speaking and Embodied Voice Coach

Share & Celebrate

MONDAY 2/1

Amplify Gallery Website goes LIVE, People's Choice voting opens

SATURDAY 2/13: AMPLIFY DAY!

10AM–NOON

Public Speaking Dress Rehearsal

1–3PM

Public Speaking Competition (plus audience trivia and epic snacks!)

3–5PM

Reception and Gallery of Op-Eds and Visual Stories and Graphics

5–6PM

Award Ceremony

Categories & Prizes

3x $500

PUBLIC SPEAKING:

Deliver a 5-minute TED-style talk on Amplify Day

3x $500

PUBLIC WRITING:

Submit an op-ed of 750-1,000 words

3x $500

WILD CARD:

VISUAL STORIES AND GRAPHICS use imagery, colors, typography, and layout to convey messages, narratives, and data rather than relying solely on text

3x $200

PEOPLE'S CHOICE

One award in each category

For More Info

If you have questions, need help getting started, or want to get on our mailing list to get all the updates, reach out to us at **wurtelecenter@smith.edu.**

Plan for Today

- Visualization Rhetoric
- Persuasive Visualization Activity

Editorial Choices

- Visualizations are created by human beings
- In that process editorial choices must be made

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- What aspects of a person's mental model of the world might influence the editorial choices they make?

- What is the difference between bias and subject matter expertise??

Rhetoric

rhetoric noun

rhet·o·ric ˈre-tə-rik

Synonyms of *rhetoric* >

1 : the art of speaking or writing effectively: such as

a : the study of principles and rules of composition formulated by critics of ancient times

b : the study of writing or speaking as a means of communication or persuasion

Visualization Rhetoric

An analytical framework for understanding how design techniques that prioritize particular interpretations in visualizations that "tell a story" can significantly affect end-user interpretation.

Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

Visualization Rhetoric

Editorial Layers

1. The Data
2. Visual Representation
3. Textual Annotations
4. Interactivity

ID one editorial decision to be made in each layer. How might that decision affect the takeaway message of the visualization?

Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

Visualization Rhetoric Techniques

Information Access Rhetoric

- Omission
  - Neglecting to cite data sources
  - Defining variables ambiguously
  - Knowledge assumptions of the end-user
  - Thresholding values
  - Omitting exceptional cases
  - Thresholding axes
  - Filtering
- Metonymy (techniques the require part-to-whole relationships)
  - Selection of variables
  - Averaging
  - Textual and visual summaries
  - Categorizing, binning, or aggregating

Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

Visualization Rhetoric Techniques

Provenance Rhetoric

- Data provenance
  - Citing and/or linking…
    - Data sources
    - Additional references
    - Methodological choices
    - Relevant facts
  - Annotating exceptions and corrections
- Representing uncertainty
  - Error bars
  - Descriptions of inferential limits (ex. CI)
  - "leap-of-faith" or forecast annotations
  - Expressions of doubt
- Identification
  - Author bios or personal anecdotes

Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

Visualization Rhetoric Techniques

Mapping Rhetoric

- Obscuring
  - Gratuitous third dimension
  - Sizing transformations
  - Discriminability limits
  - Oversizing
  - Neglect to map information to most salient visual judgement types
  - False cause-and-effect
  - Double-axis
- Visual metaphor and metonymy
  - Suggestive spatial mappings (ex. Left = past)
  - Typographical mappings
  - Color mappings
  - Visual noise

Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

Visualization Rhetoric

Mapping Rhetoric

- Contrast
  - Visual contrasts
  - Variable splices
- Classification
  - Grouping by size, position, or color
  - Consistent typographic manipulations
  - Equations of significance
- Redundancy
  - Disaggregating homogenous values or visual marks
  - Visual noise

Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

Visualization Rhetoric Techniques

Linguistic-Based Rhetoric

- Typographic emphases
  - Bolding
  - Italicizing
- Irony
  - Rhetorical questions (ex. Titles)
  - Quotation marks
  - Deliberate understatement

Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

Visualization Rhetoric Techniques

Linguistic-Based Rhetoric

- Similarity
  - Contrast
  - Analogy
  - Metaphoric statements
  - Parallelism
  - Simile
  - Double entendre
- Individualization
  - Apostrophe
  - Sorting and filtering
  - Phrasing or imagery framed from an individual-citizen level view

Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

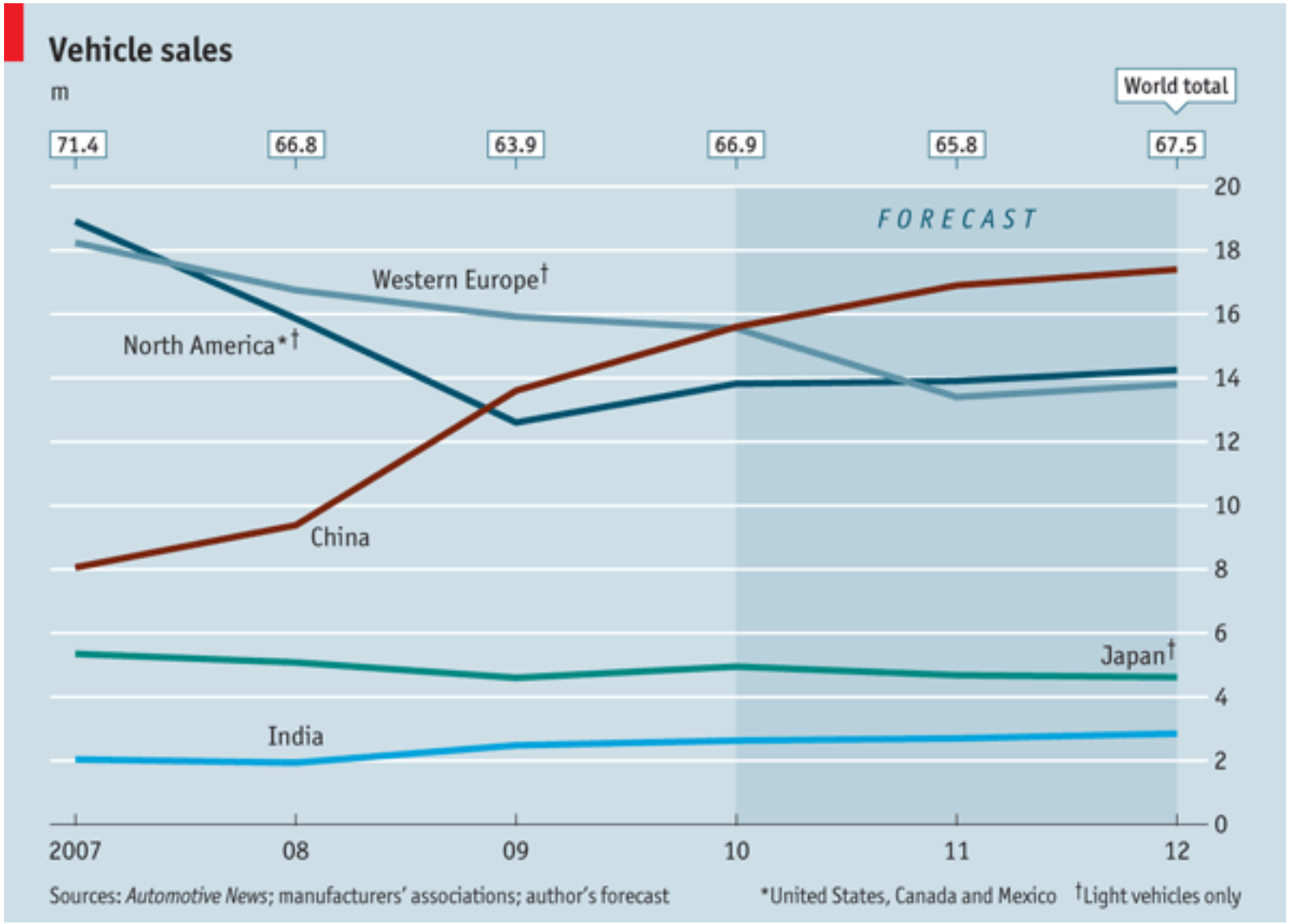
Visualization Rhetoric Techniques

Procedural Rhetoric

- Anchoring
  - Default views
  - Fixed comparisons
  - Spatial ordering
  - Animations
  - Partial animation
  - Search suggestions
  - Goal suggestions
- Filtering

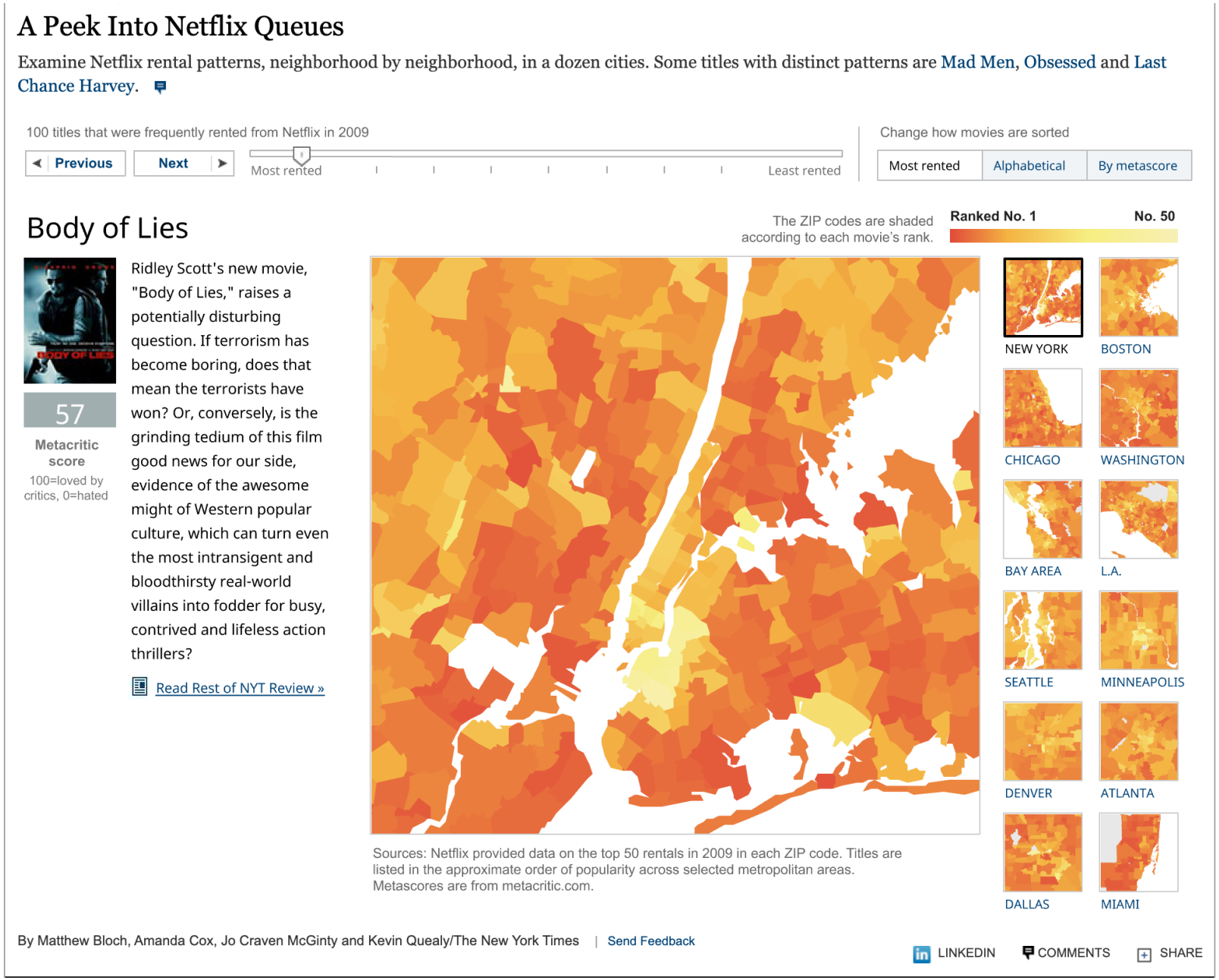
Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

What forms of Visualization Rhetoric do you see here?



Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

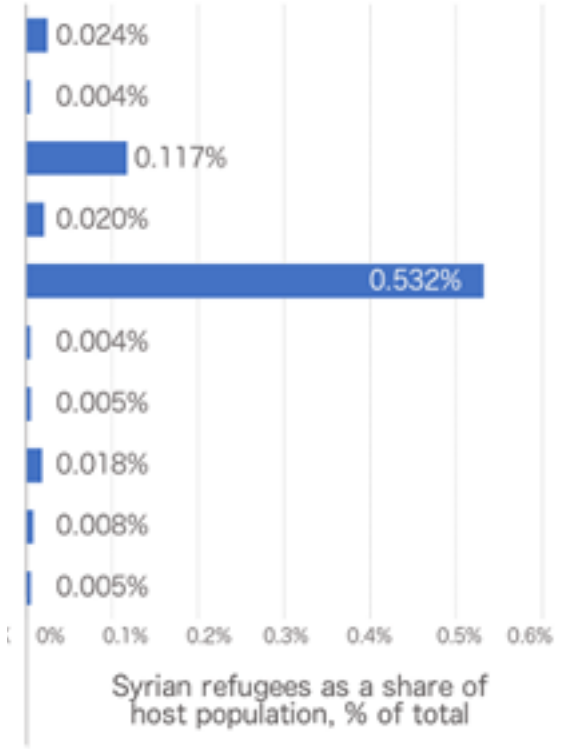
What forms of Visualization Rhetoric do you see here?



Visualization Rhetoric: Framing Effects in Narrative Visualization; Hullman and Diakopoulos

Visualization Rhetoric in Action

U.S. has accepted more Syrian refugees than Sweden, Russia, and the U.K., and more than Spain, Finland, and Australia



Sources: UNHCR Syria Regional Refugee Response
Worldometers: population by country
International Monetary Fund. Database updated on 12 April 2017

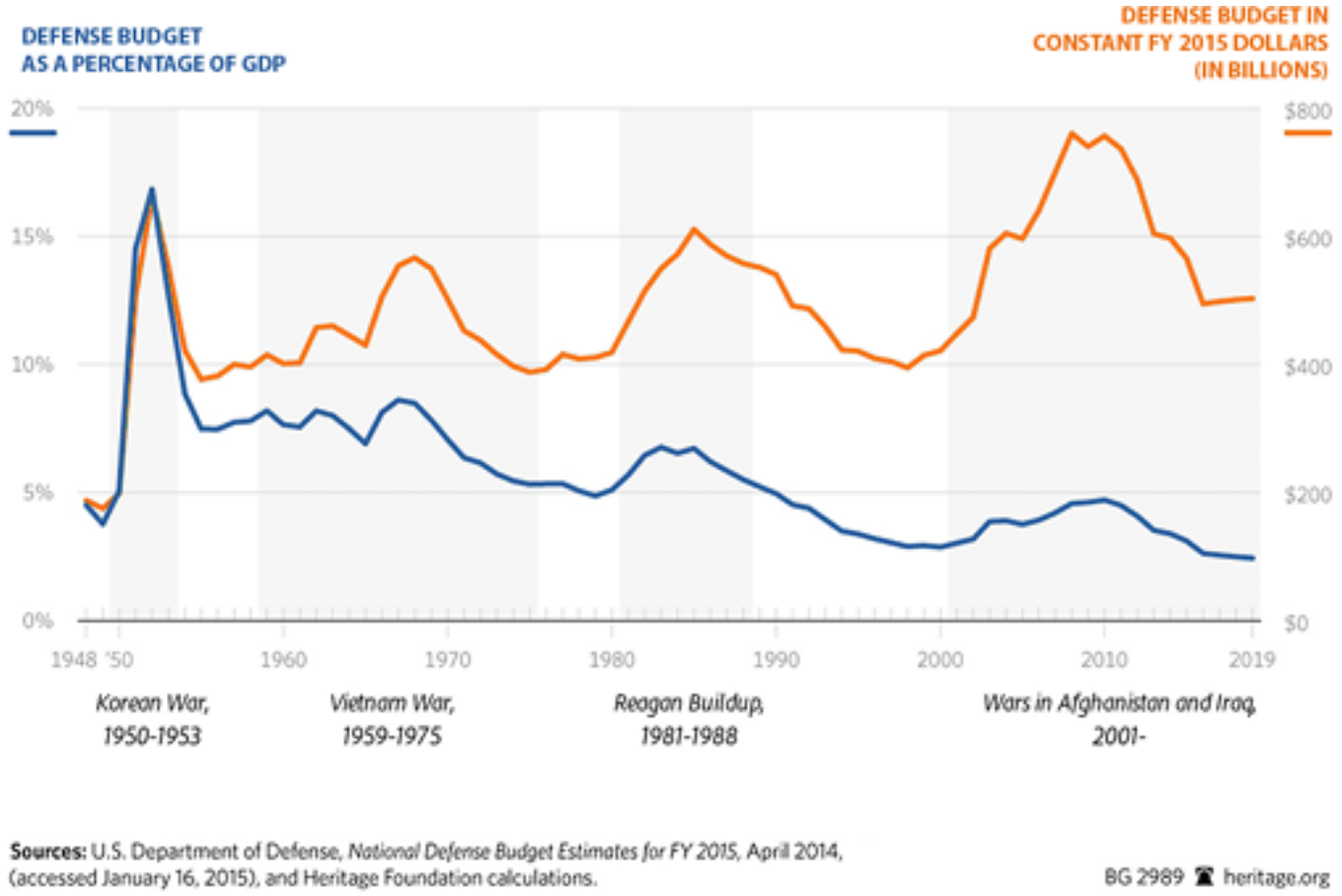
Frames and Slants in Titles of Visualizations on Controversial Topics; Kong et al.

What is the takeaway message of this visualization?

Visualization Rhetoric in Action

Defense budget on an steady decrease at a percentage of GDP over the past 50 years

Defense budget on an increase at a steady dollar amount over the past 50 years



What is the takeaway message of this visualization?

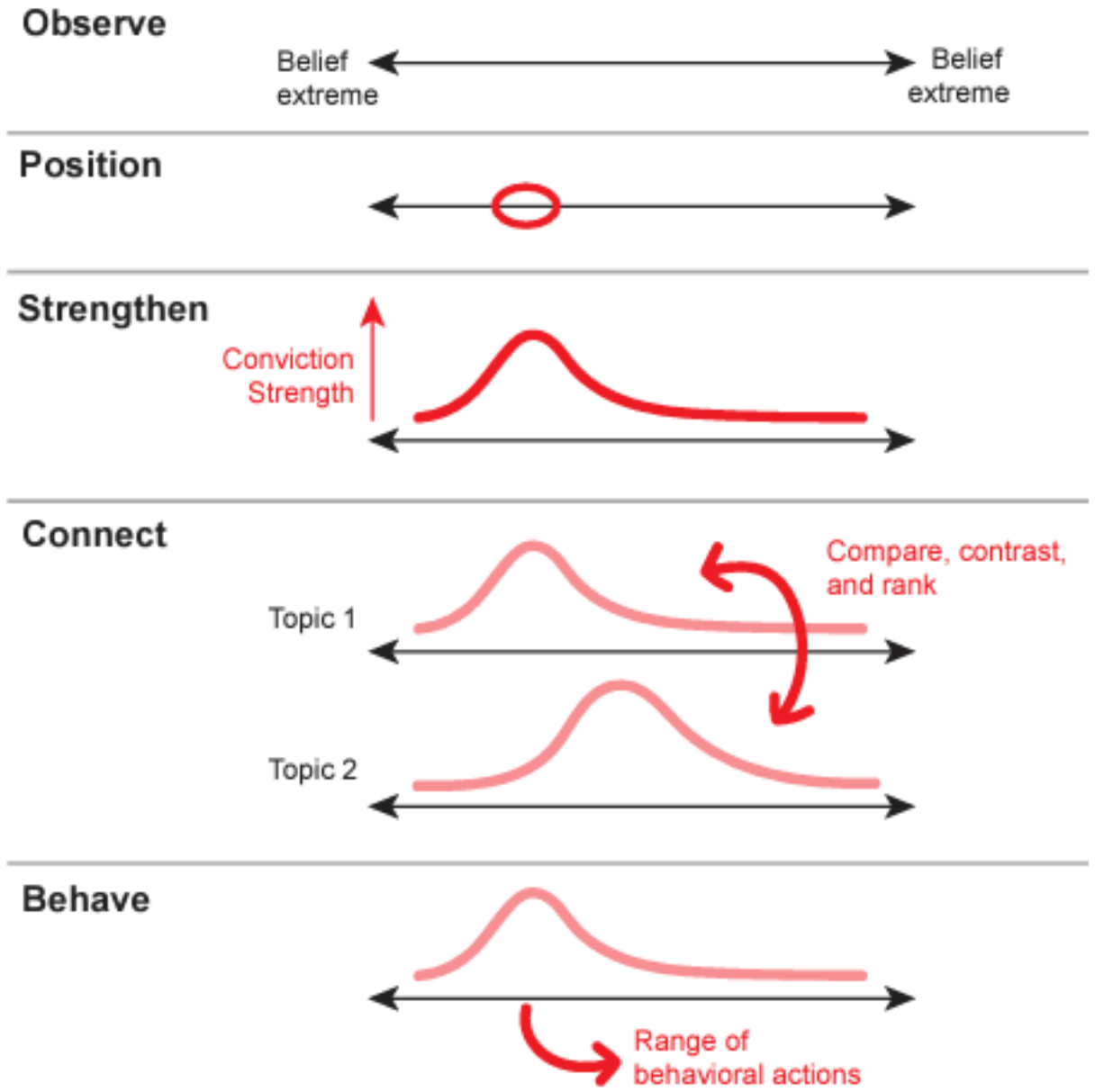
Frames and Slants in Titles of Visualizations on Controversial Topics; Kong et al.

Persuasion Techniques vs. Intent

Intent	Persuasion Technique: Logos	Persuasion Technique: Pathos	Persuasion Technique: Ethos
Cognitive	Bar chart (baseball statistics) with an objective for you to remember the data.	Visual imagery (your favorite baseball player) with an objective for you to remember the data.	Data source shown (MLB) with an objective for you to remember the data.
Affective	Bar chart (of COVID-19 deaths) with an objective for you to take action (social distance).	Visual imagery (coffins) with an objective for you to take action (social distance).	Data source shown (CDC) with an objective for you to take action (social distance).

Affective Learning Objectives for Communicative Visualizations; Lee-Robins and Adar

Affective Intents



Affective Learning Objectives for Communicative Visualizations; Lee-Robins and Adar

Impact of Metadata

Goals	Data Sources	Cleaning & Processing	Perceptual Challenges	Encoding Explanation	Creators	Intended Audience	Other
	Categories of Metadata						
Confidence in Chart	11% (13)	10% (12)	11% (13)	51% (59)	4% (4)	13% (15)	0% (0)
Confidence in Topic	14% (21)	6% (9)	7% (10)	58% (83)	1% (2)	13% (18)	1% (1)
Key Takeaways	11% (15)	8% (12)	4% (5)	65% (92)	3% (5)	9% (13)	0% (0)
Assess Trustworthiness	51% (62)	15% (18)	5% (6)	5% (6)	13% (16)	9% (11)	2% (2)
Understand Method	32% (40)	21% (27)	4% (6)	20% (25)	6% (7)	17% (21)	0% (0)
Understand Design	10% (12)	24% (27)	3% (4)	17% (19)	6% (7)	39% (45)	1% (1)
Understand Perspective	21% (25)	8% (9)	8% (9)	22% (27)	20% (24)	21% (26)	1% (1)
Satisfy Interest	26% (36)	11% (15)	7% (9)	24% (33)	7% (10)	25% (34)	1% (2)

From Invisible to Visible: Impacts of Metadata in Communicative Data Visualization; Burns et al.

Activity

- Form groups, each will be assigned one of the following datasets:
  - Human Development Indicators
  - Climate Change Indicators
  - Abortion Data by U.S. State from the Guttmacher Institute
- Explore your dataset and devise a proposition about it: a statement that asserts a judgement or opinion about the trends you might have uncovered
  - Your proposition can be contentious but ***cannot*** be dehumanizing or demeaning
- Split into two subgroups to design two persuasive visualizations; one for each side of your proposition. If you use deception, it should not be immediately obvious.
- Be prepared to share your visualizations with the class

Adapted from: https://vis-society.github.io/assignments/A4.html